HDOC Day 39

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Question 1: Weighted Average Using Variadic Function

Write a C program using double weightedAverage(int count, ...) to calculate the weighted average of value-weight pairs. Check negative weights and zero total weight, and display errors for invalid input. Use stdarg.h.

C Program

```
#include <stdio.h>
#include <stdarg.h>
#include <math.h>

double weightedAverage(int count, ...) {
    va_list args;
    double sum = 0, total = 0;
    int i;
    if (count <= 0) {
        printf("Error: Invalid count.\n");
        return NAN;
    }
    va_start(args, count);
    for (i = 0; i < count; i++) {
        double value = va_arg(args, double);
        double weight = va_arg(args, double);
        if (!isfinite(value) || !isfinite(weight) || weight < 0) {
            printf("Error: Invalid value or weight.\n");
            va_end(args);
            return NAN;
        }
        sum += value * weight;
        total += weight;
    }
    va_end(args);
    if (total == 0) {
        printf("Error: Total weight is zero.\n");
        return NAN;
    }
    return sum / total;
}

int main(void) {
    double result = weightedAverage(3, 80.0, 0.3,
                                    75.0, 0.2, 90.0, 0.5);
    if (!isnan(result))
        printf("Weighted average: %.2f\n", result);
    return 0;
}
```

Sample Output

```
Weighted average: 84.00
```

Calculation: (80 x 0.3 + 75 x 0.2 + 90 x 0.5) / (0.3 + 0.2 + 0.5) = 84.00

Question 2: Command-Line Integer Statistics

Write a C program that accepts command-line arguments, ignores invalid integers, sorts valid integers in ascending order, and displays minimum, maximum, average, median and second-largest distinct value.

Use separate functions for validation, sorting and calculation.

C Program

```
#include <stdio.h>
#include <stdlib.h>
#include <errno.h>
#include <limits.h>

int validInteger(const char *s, int *number) {
    char *end;
    long value;
    if (*s == '\0') return 0;
    errno = 0;
    value = strtol(s, &end, 10);
    if (errno == ERANGE || *end != '\0' ||
        value < INT_MIN || value > INT_MAX) return 0;
    *number = (int)value;
    return 1;
}

void sortNumbers(int a[], int n) {
    int i, j, temp;
    for (i = 0; i < n - 1; i++)
        for (j = i + 1; j < n; j++)
            if (a[i] > a[j]) {
                temp = a[i]; a[i] = a[j]; a[j] = temp;
            }
}

double average(int a[], int n) {
    double sum = 0;
    int i;
    for (i = 0; i < n; i++) sum += a[i];
    return sum / n;
}

double median(int a[], int n) {
    if (n % 2) return a[n / 2];
    return ((double)a[n / 2 - 1] + a[n / 2]) / 2;
}

int main(int argc, char *argv[]) {
    int *a, n = 0, i, second = 0, found = 0;
    if (argc < 2) {
        printf("Please provide integer arguments.\n");
        return 1;
    }
    a = malloc((argc - 1) * sizeof(int));
    if (a == NULL) {
        printf("Memory allocation failed.\n");
        return 1;
    }
    for (i = 1; i < argc; i++) {
        int value;
        if (validInteger(argv[i], &value)) a[n++] = value;
        else printf("Invalid argument ignored: %s\n", argv[i]);
    }
    if (n == 0) {
        printf("No valid integers found.\n");
        free(a);
        return 1;
    }
    sortNumbers(a, n);
    printf("Valid integers in ascending order:");
    for (i = 0; i < n; i++) printf(" %d", a[i]);
```

```
    printf("\nMinimum: %d\nMaximum: %d\n", a[0], a[n-1]);
    printf("Average: %.2f\nMedian: %.2f\n",
           average(a, n), median(a, n));
    for (i = n - 2; i >= 0; i--) {
        if (a[i] != a[n - 1]) {
            second = a[i]; found = 1; break;
        }
    }
    if (found)
        printf("Second-largest distinct value: %d\n", second);
    else
        printf("No second-largest distinct value.\n");
    free(a);
    return 0;
}
```

Sample Input (terminal)

```
gcc program.c -o program
./program 12 5 abc 8 12 20 3
```

Sample Output

```
Invalid argument ignored: abc
Valid integers in ascending order: 3 5 8 12 12 20
Minimum: 3
Maximum: 20
Average: 10.00
Median: 10.00
Second-largest distinct value: 12
```